

Aravind R

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Professional Summary

Full Stack Software Engineer (pre-final year, B.Tech AI & Data Science) specializing in **Java, Spring Boot, React.js, REST APIs, Microservices, and SQL/NoSQL databases**. Proven track record shipping and independently deploying **production-grade, cloud-hosted systems** with JWT-secured authentication, role-based access control (RBAC), OCR/AI-driven automation, and CI/CD-ready containerized architecture. Strong foundation in **DSA, OOP, DBMS, OS, Computer Networks, and System Design**, actively solving interview-level problems on LeetCode. Seeking a Software Development Engineer (SDE) / Full Stack Developer role at a product-based company.

Education

Saveetha Engineering College

B.Tech in Artificial Intelligence & Data Science; CGPA: 7.77

Chennai, India

2023 – 2027

Technical Skills

Languages: Java, Python, SQL, JavaScript, TypeScript, HTML5, CSS3

Backend & APIs: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, JDBC, RESTful APIs, Microservices, JWT Authentication & Authorization, OAuth 2.0, Maven, Swagger/OpenAPI, JUnit, Mockito, Exception Handling, DTO Pattern, Repository Pattern

Frontend: React.js, React Hooks, Redux, Axios, Bootstrap, Tailwind CSS, Responsive Web Design, Next.js

Databases: MySQL, PostgreSQL, Oracle SQL, H2, Redis, MongoDB, Database Design, Normalization, Indexing, Query Optimization
Cloud, DevOps & Tools: Docker, Docker Compose, Kubernetes, Git, GitHub, GitHub Actions, Jenkins, CI/CD, AWS (EC2, S3, RDS, IAM), Linux, Postman, IntelliJ IDEA, VS Code, Jira, Terraform

Messaging & Distributed Systems: Apache Kafka, RabbitMQ, WebSocket, Event-Driven Architecture, REST, gRPC

Architecture & Design: System Design, Design Patterns, SOLID Principles, Clean Architecture, MVC Architecture

Core Computer Science: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Multithreading, Concurrency, Agile/Scrum

AI/ML: NLP, OCR, TF-IDF, Cosine Similarity, Streamlit

Work Experience

Full Stack Developer Intern

ApproTech

Jul 2025

Chennai, India

- Designed and deployed **5+ RESTful API endpoints** using Java and Spring Boot; optimized MySQL queries, reducing backend response time by **~30%**.
- Implemented **Spring Security with JWT authentication**, securing all API endpoints and eliminating unauthorized access in the test environment.
- Built **3 reusable React.js components** with Hooks and Axios integration for seamless frontend-backend data flow.
- Performed end-to-end **API testing using Postman** across 10+ endpoints and collaborated in 2-week **Agile sprint** cycles with consistent on-time feature delivery.

Projects

End-to-End Document Intelligence & Decision Automation System (DocFlow)

Java, Spring Boot, JWT, MySQL, OCR, React.js, Docker

GitHub | Live Demo

- Configurable workflow automation platform** (not a static document manager) — business rules are admin-configurable rather than hardcoded, so organizations adapt the platform to new processes by editing rules, not code.
- On upload, content is extracted via **OCR** and passed to a **Rule Engine** that matches it against admin-defined rules to auto-determine department, priority, workflow category, and decision; unmatched documents route automatically to **manual review**.
- Built a three-tier **APPROVE / REVIEW / REJECT** pipeline across **200+ document types** with weighted keyword-confidence scoring, cutting processing time by **~50%** versus manual workflows.
- Implemented **JWT-secured REST APIs with RBAC** (Admin, Staff, Viewer), notifications, analytics, audit logging, document preview, and reporting; containerized with **Docker** across 22+ production deployments.
- Independently deployed full production infrastructure: Spring Boot on **Render**, React on **Vercel**, MySQL on **Railway**, with automated **UptimeRobot** health checks for continuous uptime.

Own Workspace — Multi-Tenant Productivity Suite (Mail, Chat, Docs, Sheets)

GitHub* | Live Demo*

Spring Boot, React.js, PostgreSQL, Redis, WebSocket, Docker, JWT

- Architected a **Google Workspace/Microsoft 365-style** multi-tenant SaaS suite unifying an own **Mail** client, **Chat** app, **Docs** editor, and **Sheets** module under one workspace login, with **tenant.id** row-level data isolation and per-tenant subdomains.
- Built an own **Mail module** (compose/send/receive, threaded inbox, folders, search) and an own **WhatsApp-style Chat module** with WebSocket-powered real-time group messaging, presence indicators, and read receipts.
- Developed an own **Docs module** (Word-style rich-text editor with real-time multi-user collaborative editing) and an own **Sheets module** (Excel-style grid with formula evaluation and cell dependency computation).
- Implemented a **5-level RBAC hierarchy** (Super Admin, Tenant Admin, Manager, Member, Viewer) with **JWT + refresh tokens**, per-tenant **Redis token-bucket rate limiting**, subscription/feature-gating middleware, and an immutable audit-log system.
- Containerized all modules (Mail, Chat, Docs, Sheets) with **Docker Compose** for independent, horizontally scalable services.

Banking Management System

GitHub | Live Demo

Java, Spring Boot, Spring Security, JWT, React.js, MySQL, REST APIs

- Built a full-stack banking system with **JWT-secured authentication** for account management, fund transfers, deposits, withdrawals, and complete **transaction tracking**.
- Designed RESTful **Spring Boot APIs** integrated with a **React.js** frontend and **MySQL** database for persistent, auditable transaction management.

AI Resume Analyzer Web Application

GitHub

Python, NLP, TF-IDF, Cosine Similarity, Streamlit

- Built a resume-analysis app using Python/NLP with **TF-IDF vectorization** and **cosine similarity** scoring to match resumes against job descriptions, validated across **50+ test resumes**.
- Automated keyword and structure analysis, surfacing missing skills and improvement suggestions, reducing manual resume-screening effort by **~40%**; deployed via an interactive **Streamlit** interface.

*Temporary placeholder link — replace with dedicated repo/live demo once deployed.

Certifications

Oracle SQL Certified Specialist • NPTEL – IoT (Elite) • PyTorch Ultimate (Coursera) • Microsoft Azure Learning Challenge • JPMorgan Chase – Software Engineering Virtual Experience (Forage) • IBM – Software Engineering

Achievements

Participated in the **Google Gen AI Exchange Hackathon** (national-level, Google Cloud & Hack2Skill) — built and presented an AI-driven prototype under competitive time constraints.

Participated in the **Hack Hustle Hackathon**, collaborating on real-world problem-solving solutions.